

SVS GROUP OF INSTITUTIONS

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MINI PROJECT

LANGUAGE TRANSLATOR

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OBJECTIVE

- To provide an overview of language translator technology and its impact on communication.
- To explore the implementation and the benefits of using a language translator



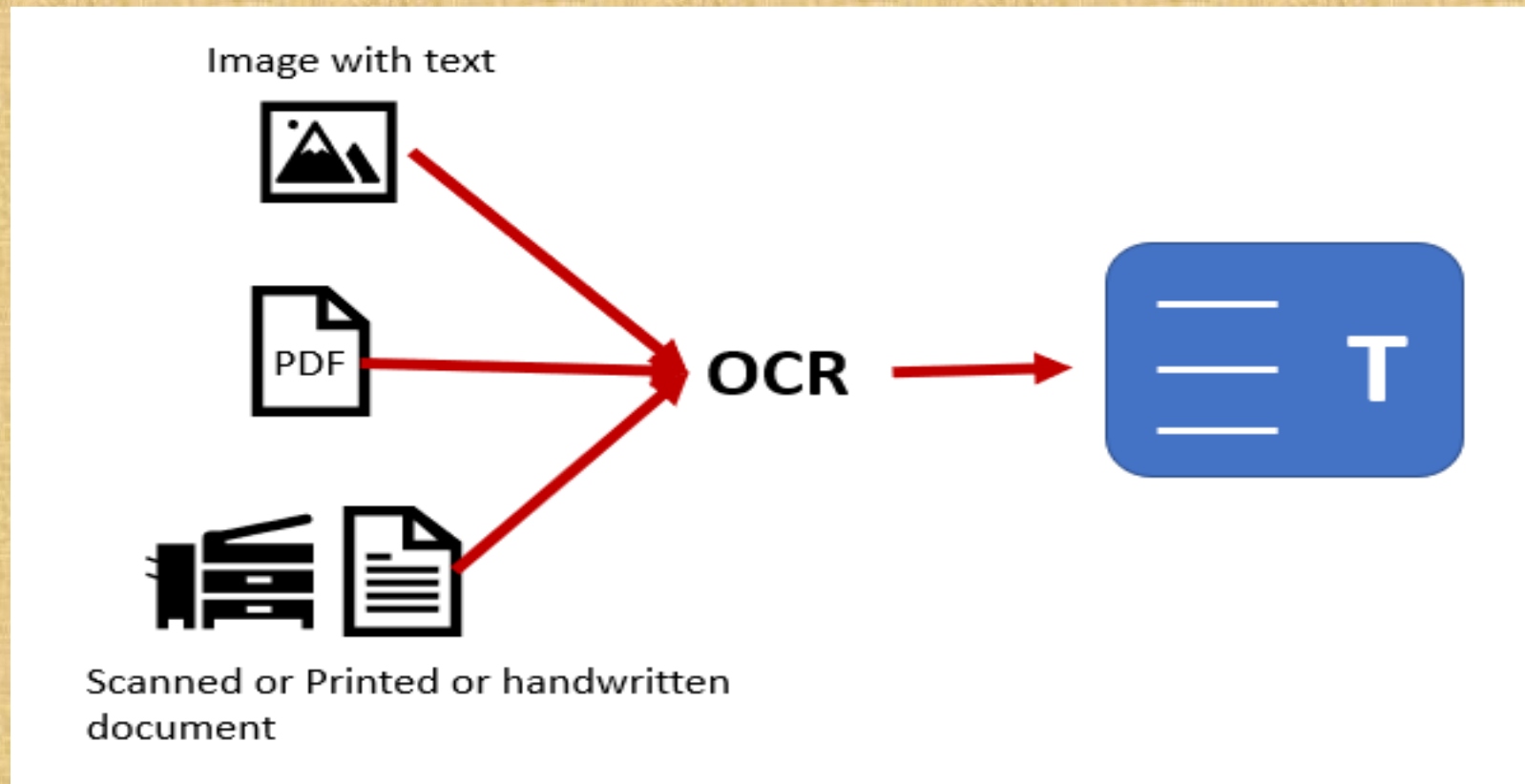
WHAT IS LANGUAGE TRANSLATOR ?

- ▶ A language translator is a software program designed to translate text, speech or written content from one language to another.
- ▶ Language translators are widely used in various fields to facilitate communication between people who speak different languages.
- ▶ Language translators use various algorithms and Techniques to analyze and understand the:
 - a) Context
 - b) Grammar
 - c) Syntax of the language and then translates to other Target language accurately.



OPTICAL CHARACTER RECOGNITION(OCR)

- OCR technology is designed to recognize and extract characters, words, and sentences from a scanned or photographed image of a document.



OPTICAL CHARACTER RECOGNITION (CONTD.)

- Python-tesseract is an optical character recognition (OCR) tool for Python. That is, it will recognize and “read” the text embedded in images.



REQUIREMENTS

Hardware_Requirements:

Processor : Min. Core i5 processor
RAM : 2GB(Min) or 4GB(Recommended)
Hard Disk Space : 20GB+

Software_Requirements:

Operating system : Windows 7 or latest version

1. OCR : pip install pytesseract (for extracting the characters from uploaded image).
2. langdetect : pip install langdetect (To detect the language of text extracted).
3. translate : – pip install translate (To convert the extracted text to the target language).



METHODOLOGY

- Upload the captured image as input.
- Extract the text from the image.
- Detect the language of the extracted text
- Take the target language as input from the user
- Check whether the target language is the same as the detected language. Tell the user that the extracted language is the same as the given language by you. And ask them to re-enter another language.
- Generate the converted language as output.



EXPECTED OUTCOME

Enter the name of your text file (e.g., 'deep.txt'): C:\Users\KAMTAM DEEPAK KUMAR\Pictures\Screenshots\New folder\dam.txt

🌐 Detected Language: en

Enter target language code (e.g., 'hi' for Hindi, 'fr' for French): hi

✅ Translated Text:

भारत, अधिकारिक तौर पर भारत गणराज्य, [j] [20] दक्षिण एशिया में एक देश है। यह क्षेत्रफल के हिसाब से सातवां सबसे बड़ा देश है; 2023 के बाद से सबसे अधिक आबादी वाला देश; [21] और, 1947 में अपनी आजादी के बाद से, दुनिया का सबसे अधिक आबादी वाला लोकतंत्र। [22] [23] [24] दक्षिण में हिंद महासागर, दक्षिण-पश्चिम में अरब सागर और दक्षिण-पूर्व में बंगाल की खाड़ी से घिरा हुआ यह पश्चिम में पाकिस्तान के साथ भूमि सीमा साझा करता है; [k] उत्तर में चीन, नेपाल और भूटान; और पूर्व में बांग्लादेश और म्यांमार।



CONCLUSION

- Language translators used in different fields such as health care, government, industries, education, business, and tourists to communicate with people speaking different foreign languages.
- Language translator technology has significantly impacted communication and language barriers across the globe.
- Language translator technology will continue to evolve and improve, making it an essential tool for global communication in the future.



REFERENCE

- 1 Pramudita, Y. D., Putro, S. S., Wahyudi, R. N., Suzanti, I. O., & Solihin, F. (2020). RESTful Web Service for Madurese and Indonesian Language Translator Applications on Android Devices. 2020 6th Information Technology International Seminar (ITIS). doi:10.1109/itis50118.2020.9320992.
- 2 Priya, L., Sathya, A., & Raja, S. K. S. (2020). Indian and English Language to Sign Language Translator- an Automated Portable Two Way Communicator for Bridging Normal and Deprived Ones. 2020 International Conference on Power, Energy, Control and Transmission Systems (ICPECTS). doi:10.1109/icpects49113.2020.93





THANK YOU!

